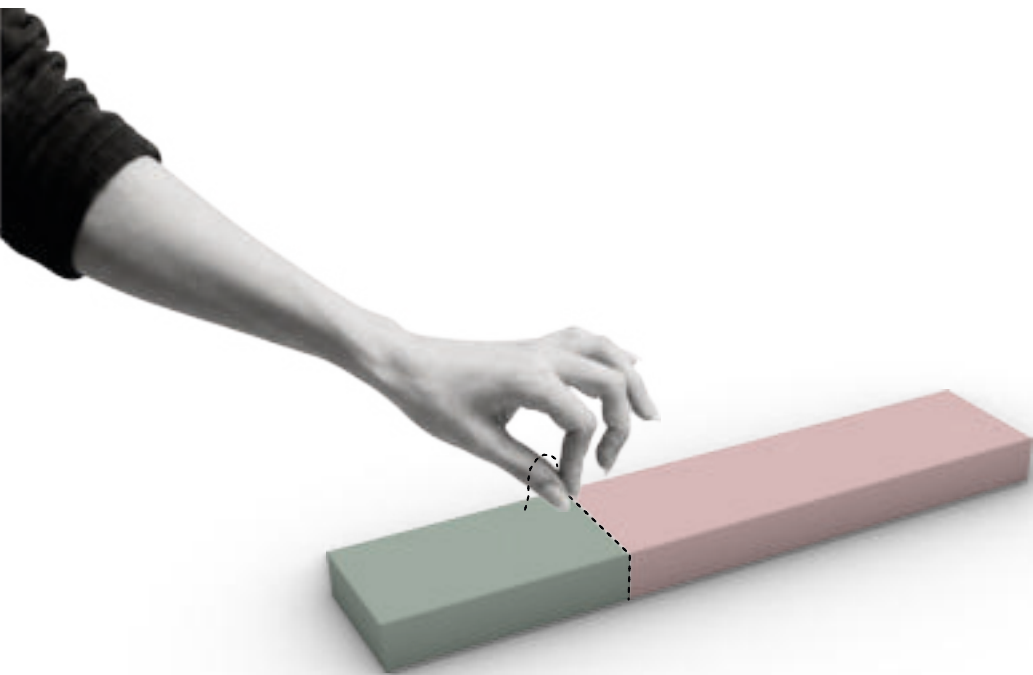


Elastic logistics

Rethinking Logistics



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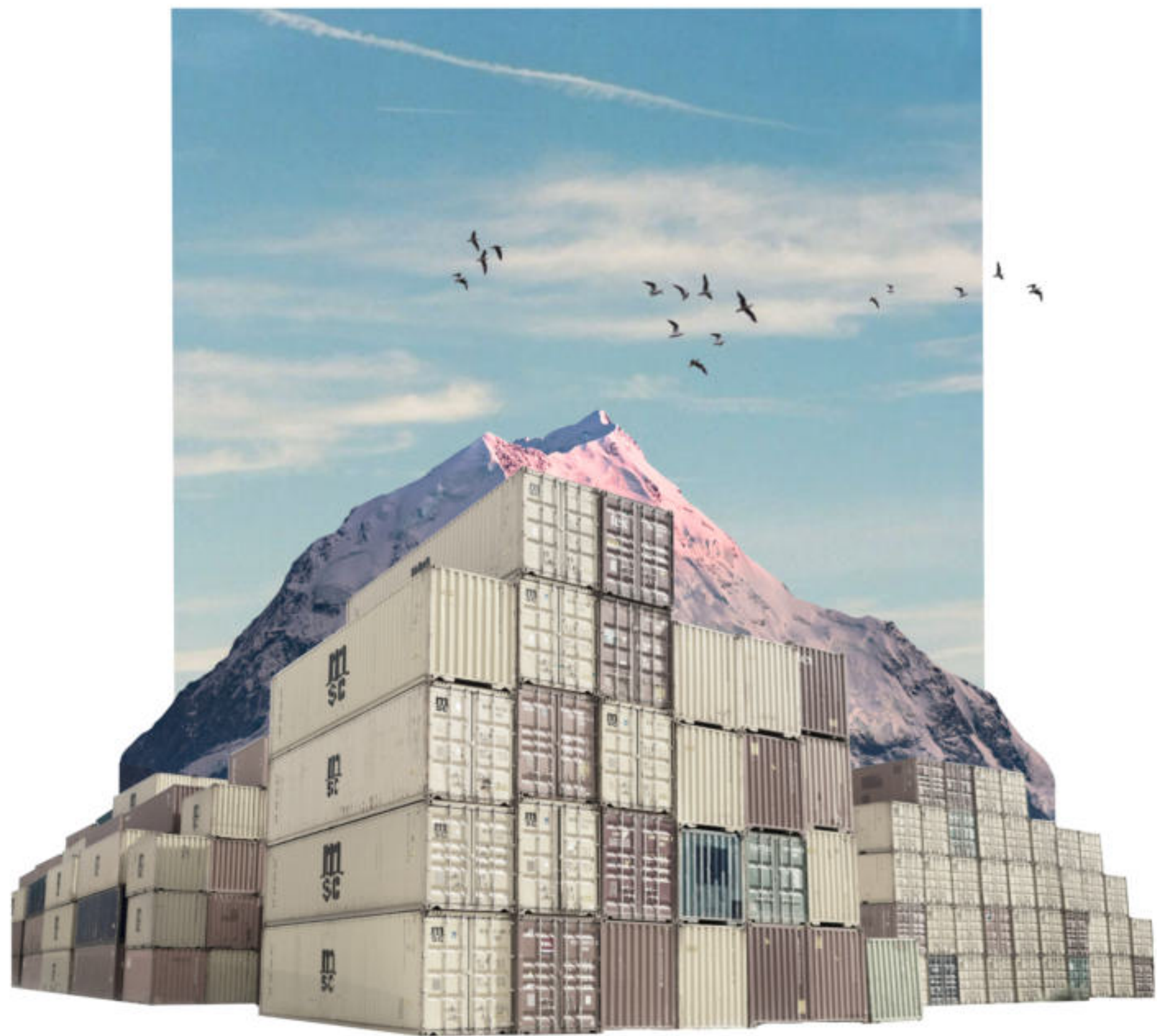
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Studio brief

This studio looks at logistics in a new way. Instead of seeing it as something hidden in the background, we treat it as an important part of everyday life that shapes how cities function. From deliveries to storage to movement, logistics is everywhere , we just rarely notice it.

The goal is to question how these systems work today and imagine how they could work better. How can logistics be more sustainable? more efficient? more connected to people and the city? Rather than keeping these spaces purely industrial, we explore ways to design them so they become more sustainable, flexible, and even enjoyable parts of the built environment.

This studio was really driven by research and evidence-based decisions. Instead of jumping straight into designing, we spent a lot of time actually understanding how logistics works today, the systems, the data, and all the constraints behind it, and letting that shape the design direction.

In collaboration with Henning Larsen and Panattoni, along with Ramboll, the Urban Land Institute (ULI), and different logistics experts. Having all these perspectives made a big difference, because it constantly grounded the work in reality and pushed it beyond just a conceptual exercise.

Case study



Sketch of existing building

In the realm of Icelandic urban planning, the term "*Græna gímaldið*" (**The Green Monstrosity**) has quickly transformed from a local nickname into a cautionary planning concept. Located in Breiðholt, a neighbourhood of Reykjavík, this monolithic steel-frame warehouse has sparked severe community backlash, widespread media attention, and ongoing legal battles.

In 2025, it was voted the "**Ugliest New Building in Iceland**" by *Arkitektúruppreisnin* (The Architectural Uprising), capturing over 43% of the public vote.



Photographs of existing building



Elevation of existing building

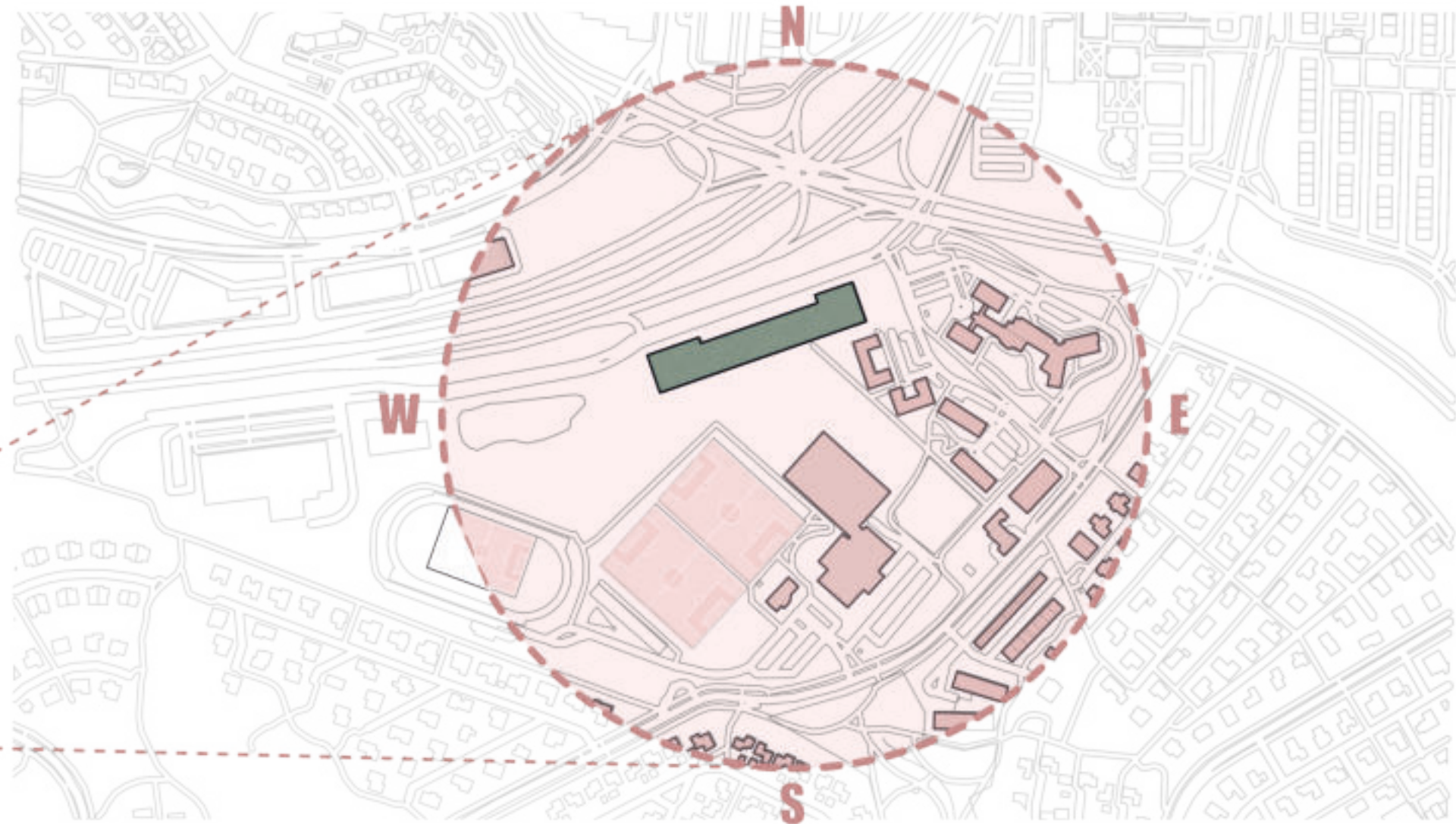
Location



Iceland



Reykjavik



Logistics hub

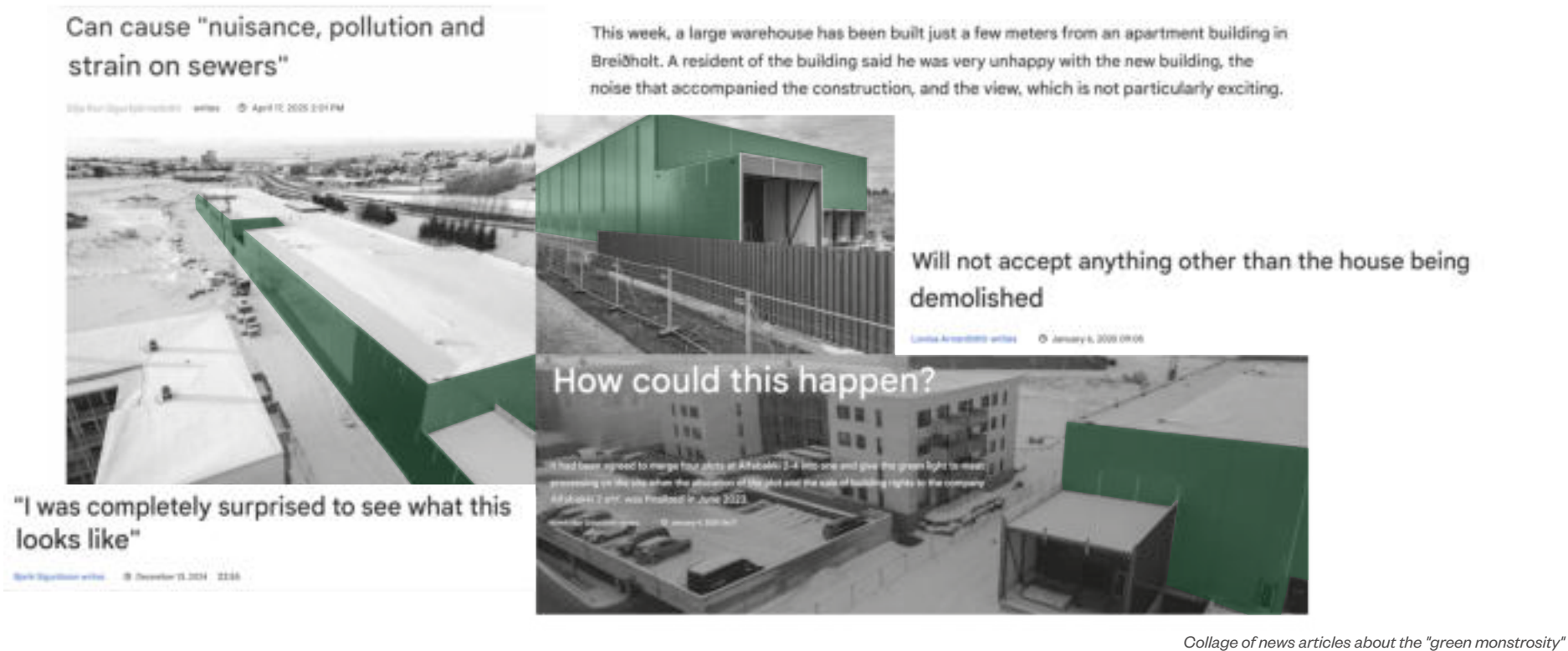


Residential



Comercial

How could this happen ?



The Neighborhood Plan Goal: The Reykjavík Municipal Plan 2040 outlines clear goals for Breiðholt: urban repair, walkability, a mix of services, and creating an inclusive, high-quality built environment. It explicitly designates this zone as a high-density mixed-use area.

The Reality of the "Monstrosity": Instead of a vibrant, street-facing commercial hub, the city approved a massive steel-frame logistics warehouse (11,000m² footprint initially, later expanded to over 14,000m²). It houses operations for a major supermarket chain, including meat processing and food packaging.

The Planning Failure: The planning department justified this by claiming it fell under "commercial use." However, in practice, the building's scale, blank green walls, and lack of ground-floor interaction act as heavy industrial infrastructure directly abutting high-density residential properties.

Downgrading the quality of the neighbourhood: Located just 14 meters from the nearest residential building, the "Monstrosity" truly lives up to its name, obstructing the view from every apartment in the building. Its overwhelming scale drastically reduces natural daylight, effectively turning even third-floor apartments into dark, basement-like living spaces with little to no direct sunlight reaching the windows.

What are logistics like today?

Today's logistics landscape is still dominated by the traditional "big box" warehouse; **large, rigid structures** that no longer fit the fast-moving world we live in. These spaces were designed for stability and permanence, but today's reality is constantly changing. Consumer demand shifts unpredictably, cities are growing denser, and supply chains are under pressure to move faster than ever before. Yet our logistics infrastructure remains **static, slow to adapt, and environmentally demanding.**

This rigidity creates a series of challenges. Urban areas become congested with delivery trucks moving between oversized distribution hubs, while warehouses themselves quickly become overcrowded and inefficient. Large amounts of land and energy are consumed even when these spaces are underused, resulting in unnecessarily high carbon footprints. At the same time, these environments are designed purely for storage and efficiency, often overlooking the human experience and offering little flexibility or value to the surrounding community.

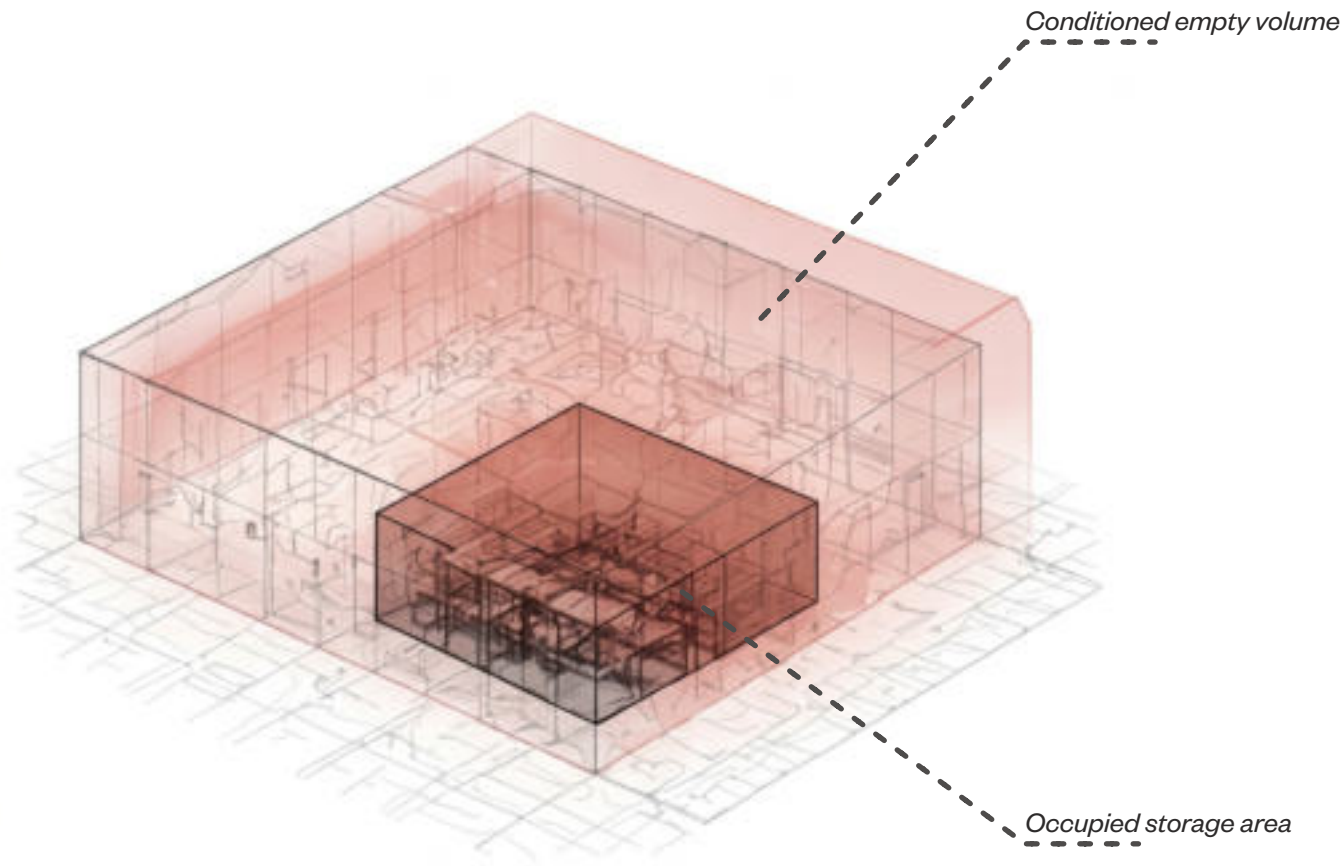
We are living in a time of constant urgency and movement, but our logistics architecture is still rooted in outdated ideas of permanence. To move forward, we need to rethink the warehouse not as a fixed box, but as a **flexible and responsive system**, one that can evolve alongside the changing rhythms of the world it serves.



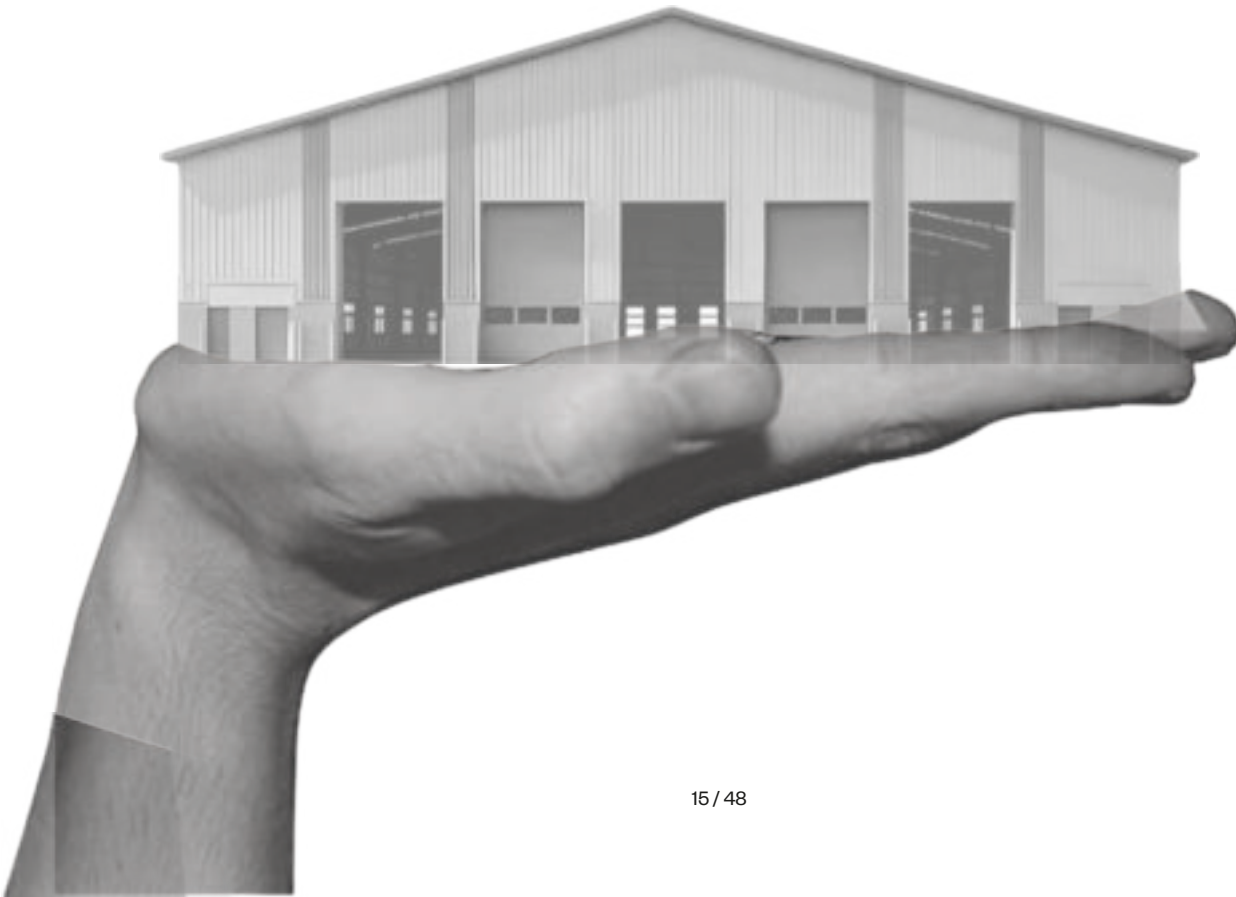
The Failure of the Big Box

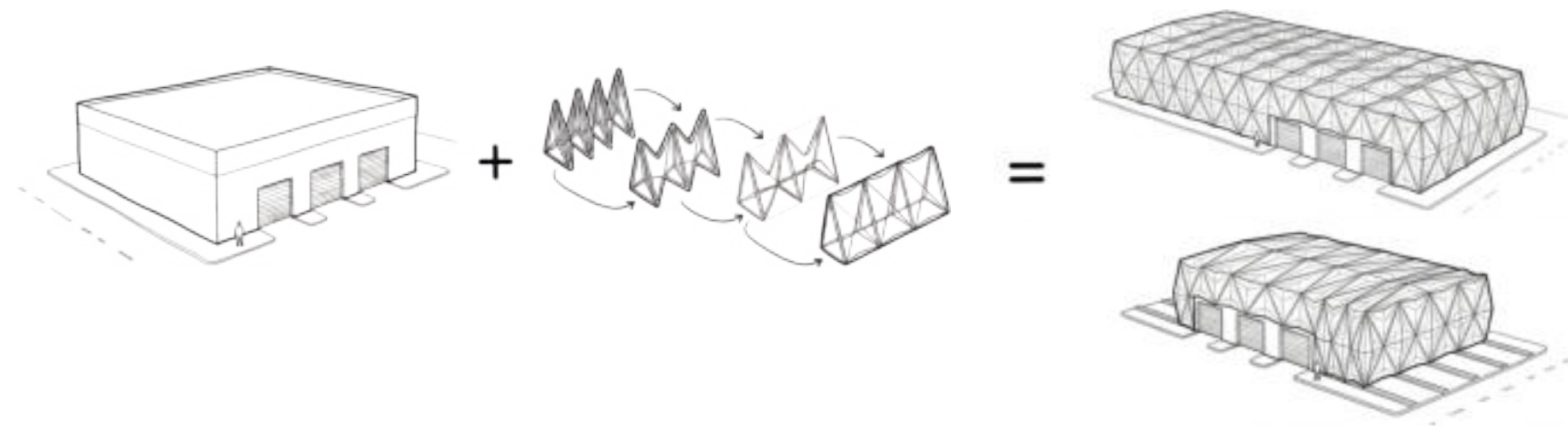
Contemporary logistics architecture is dominated by the rigid “Big Box” warehouse; oversized structures designed for permanence rather than adaptability. Built for peak demand, these spaces often remain partially empty while still requiring continuous heating, ventilation, and maintenance.

In Iceland’s harsh climate, lightweight steel envelopes perform poorly against extreme wind and temperature conditions, resulting in significant thermal loss and unnecessary energy consumption. Beyond their environmental impact, these monolithic buildings create urban dead zones; disconnected from public life and incapable of adapting to the changing rhythms of the city.



How can we change this?





Concept

Elastic Logistics is a flexible, adaptive architecture inspired by deployable systems and responsive infrastructure. It challenges the inefficiency of the traditional static warehouse by introducing a building capable of physically expanding, contracting, and reorganizing itself according to demand.

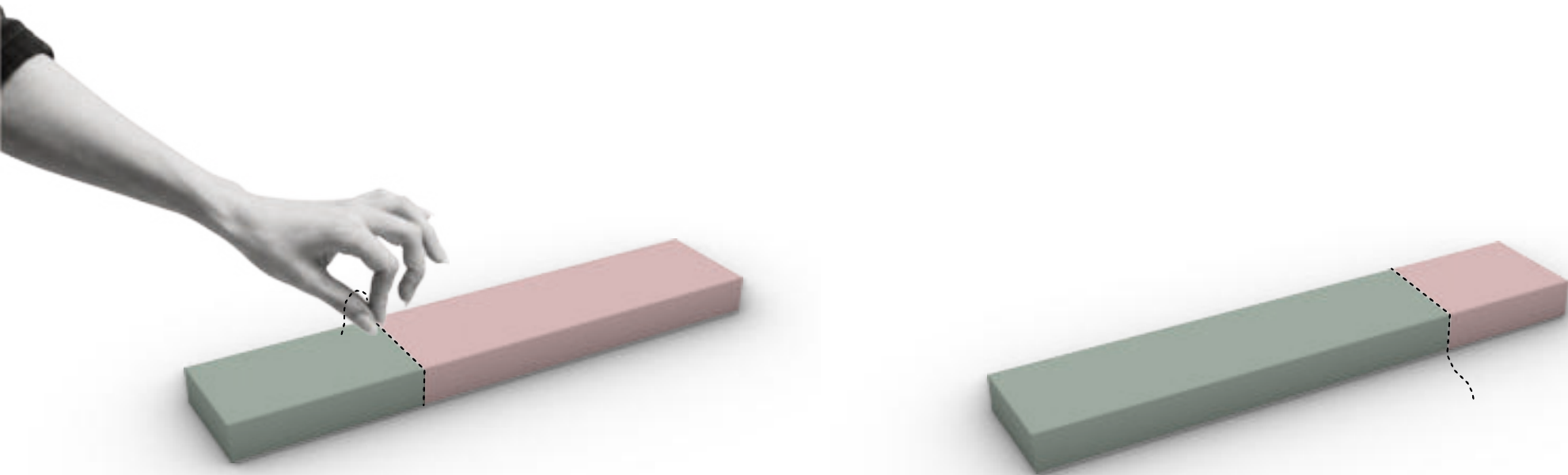
The project explores what happens when architecture stops being permanent and begins reacting; when a building no longer simply occupies space, but continuously adapts to it. Through shifting spatial zones, deployable structures, and responsive thermal boundaries, the architecture operates almost as if it is breathing.

Elastic Logistics moves between opposites: private and civic, opaque and transparent, storage and public experience. By synchronizing spatial capacity with real-time demand, the project reduces urban spatial waste while allowing the building to remain an active asset to the city throughout changing seasons and occupancy conditions.



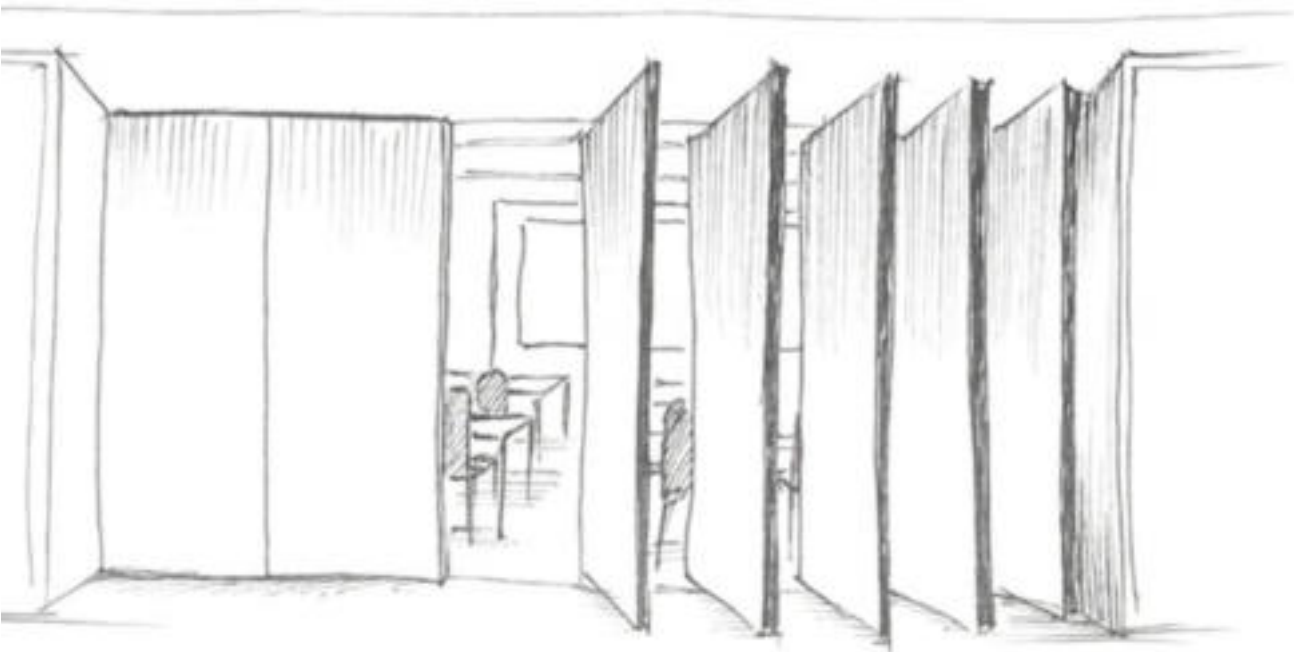
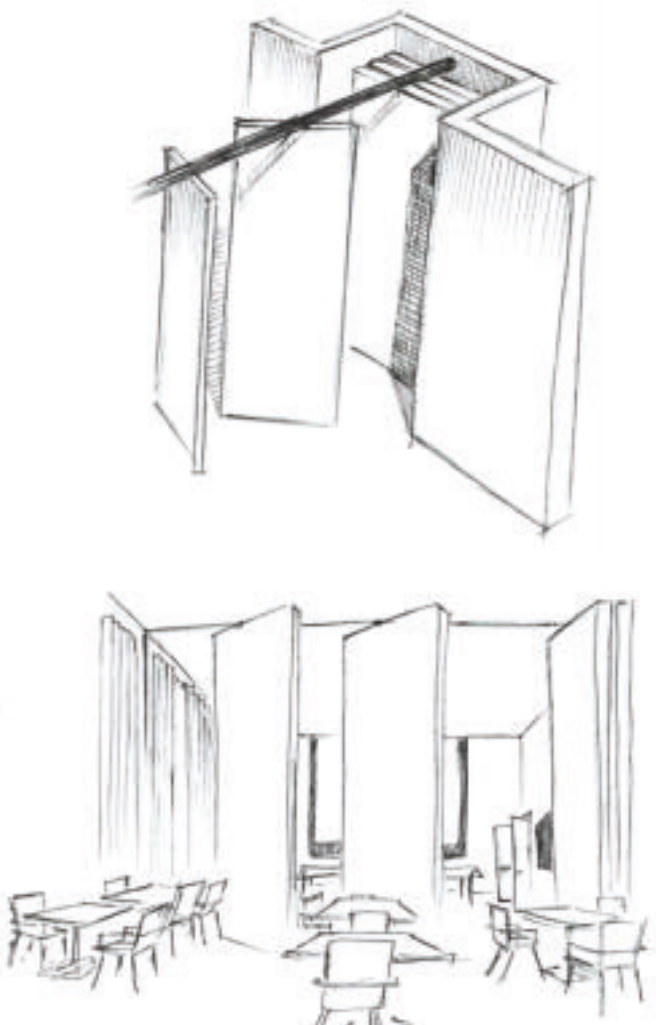
Adaptive Partition Systems

The core of the elastic floor plan lies in its ability to physically reallocate space according to real-time inventory demand. Instead of maintaining a fixed ratio between logistics, administrative, and public functions, the building operates through a system of movable partition walls capable of expanding or compressing active storage zones.



Traditional logistics buildings are designed for peak seasonal capacity; conditions reached only during short periods such as Black Friday, Christmas, or major import cycles. For much of the year, large portions of these warehouses remain underused while still requiring continuous heating, lighting, and maintenance.

In Reykjavík's cold climate, this results in significant amounts of energy being spent conditioning inactive air volume. By shrinking or expanding the active thermal zone, the project reduces unnecessary heat loss and redefines spatial efficiency as a form of energy efficiency.

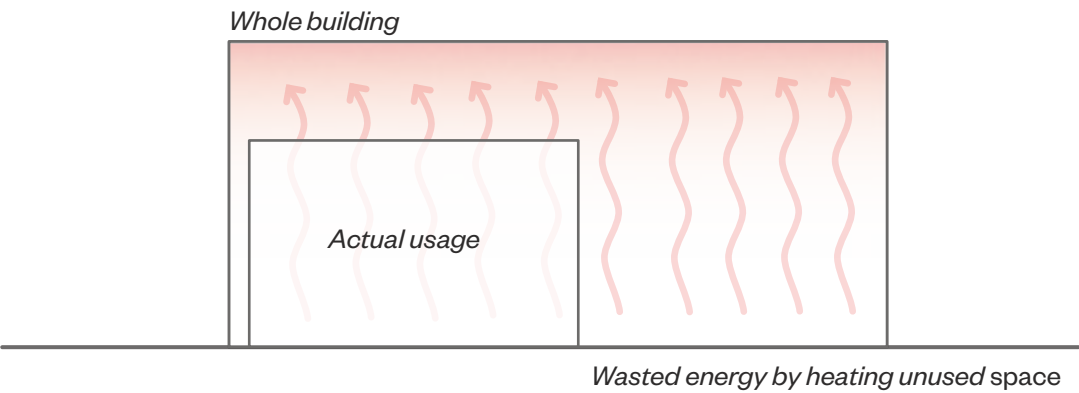


Looking at a warehouse building in Reykjavík, Iceland, a typical energy consumption is approximately 100 kWh/m²/year.

Looking at a simple 6.000m² building, that means the energy consumption is

600,000 kWh

per year for the whole building.

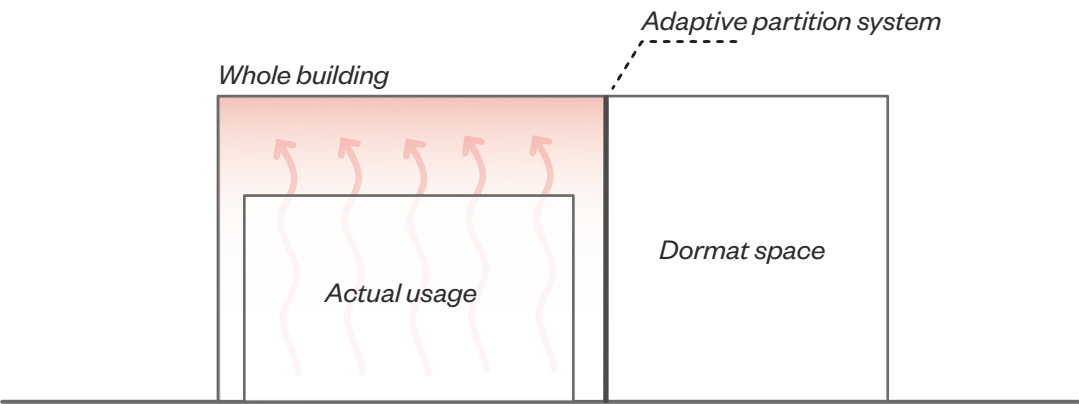


By reducing the active thermal zone to 60% during off-peak periods, the operational heating demand can be significantly reduced:

Active thermal zone is reduced to 3.600m² building, that means that the energy consumption will be around

360,000 kWh

per year for the whole building.



This creates an estimated annual **reduction** of:

240,000 kWh/ year

Energy calculations are conceptual estimates used to compare spatial heating strategies.

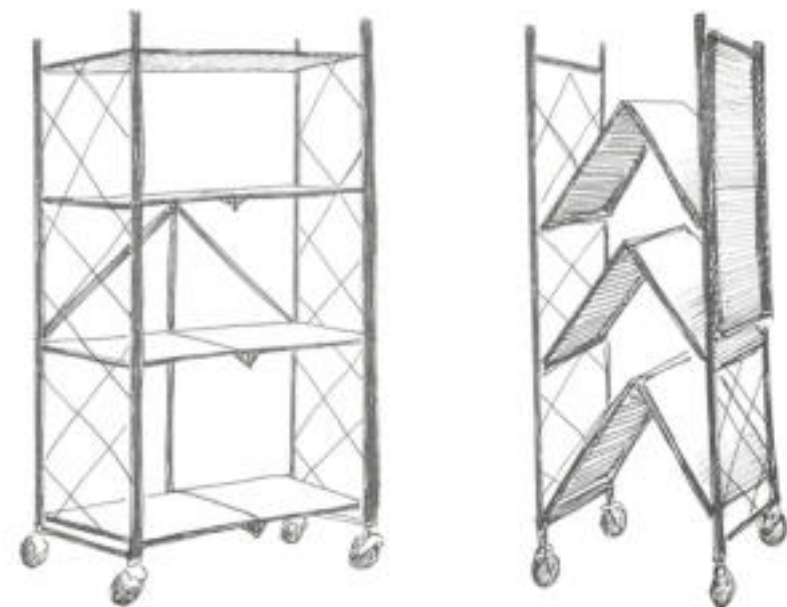
As logistics areas contract, dormant spaces can temporarily transform into public programs such as galleries, event spaces, or civic interiors. These environments become partially **activated through human occupancy** itself, reducing reliance on permanent mechanical conditioning while allowing the building to remain active throughout changing seasonal demands.



To achieve true spatial elasticity, the internal infrastructure must be as flexible as the building envelope. The project utilizes a foldable, modular racking system designed for rapid reconfiguration.

Traditional fixed racks lock a building into a permanent industrial layout. A foldable system allows the "dormant" side of the building to be cleared entirely, transforming the warehouse floor into a completely open, high-volume hall suitable for art exhibitions and public events.

The racks are designed on a manual telescopic or "scissor" mechanism, allowing staff to compress storage capacity without the need for heavy machinery or dismantling. This ensures that the building can respond to supply chain changes in a matter of hours, rather than days.



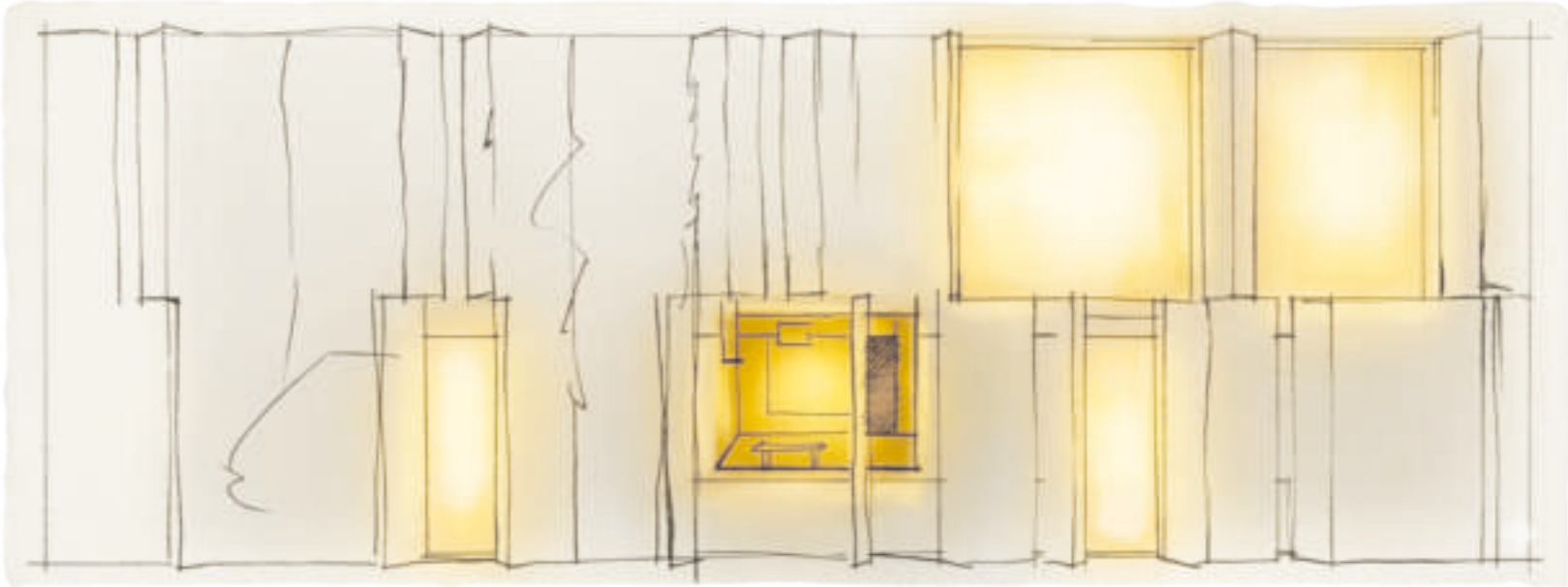
Elastic Boundaries

The transformable facade allows the building to shift between two spatial identities according to operational demand.

During low-demand periods, the envelope opens to bring daylight deep into the interior while strengthening the relationship between the building and the public realm. Dormant logistics zones can temporarily transform into flexible civic spaces, allowing the warehouse to operate as part of the urban fabric rather than as a closed industrial object.

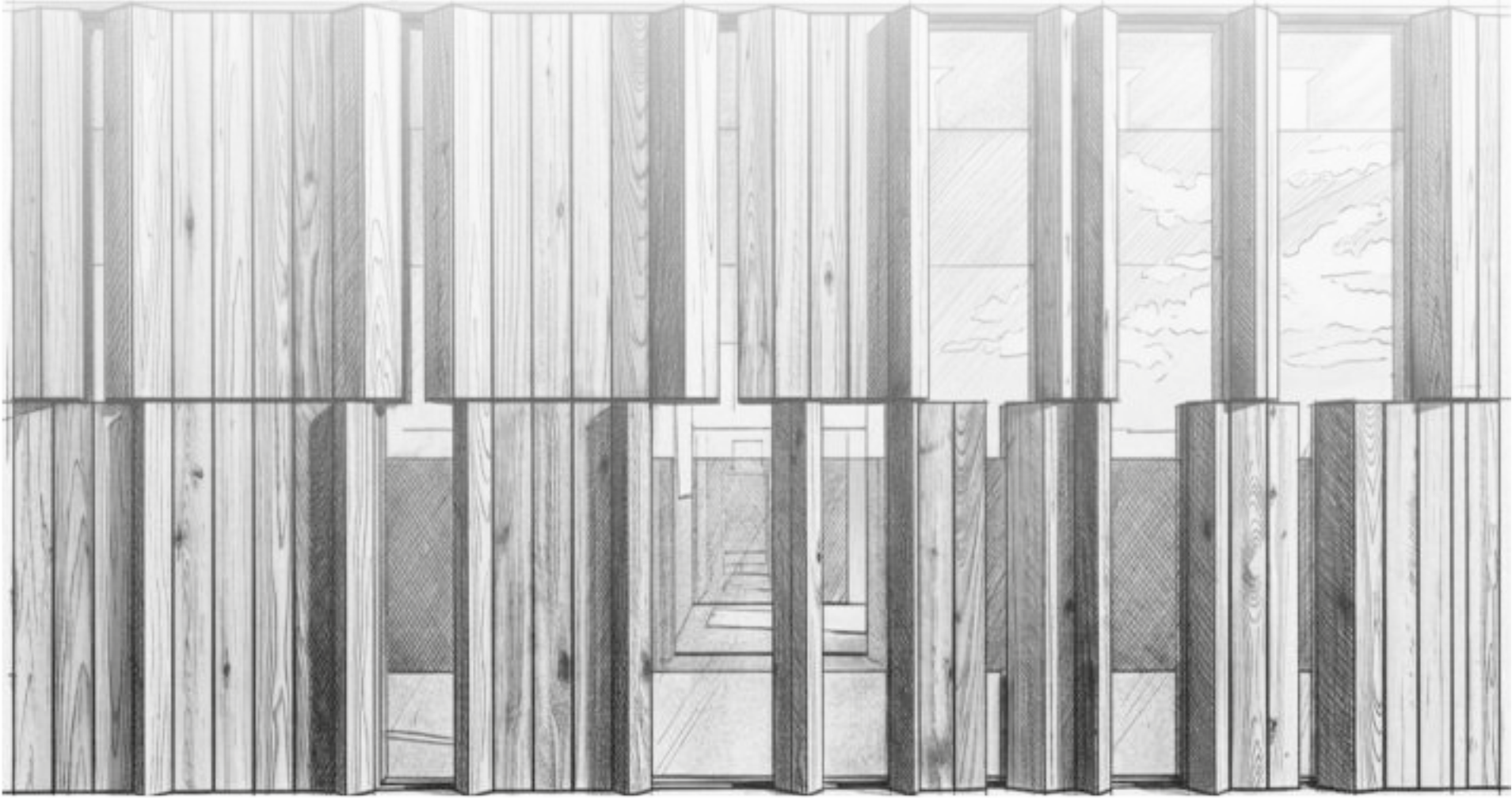
During seasonal demand peaks, the facade contracts into a compact and thermally efficient configuration optimized for logistics operations. Reduced openings minimize unnecessary heat loss while creating a controlled environment focused on storage efficiency, inventory protection, and operational performance.

The facade therefore operates as both an **environmental and spatial system**, continuously adapting between public openness and industrial efficiency throughout changing seasonal conditions.

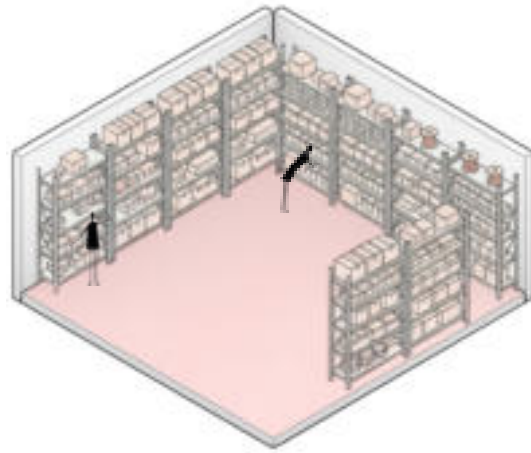




Facade system closed



Facade system open



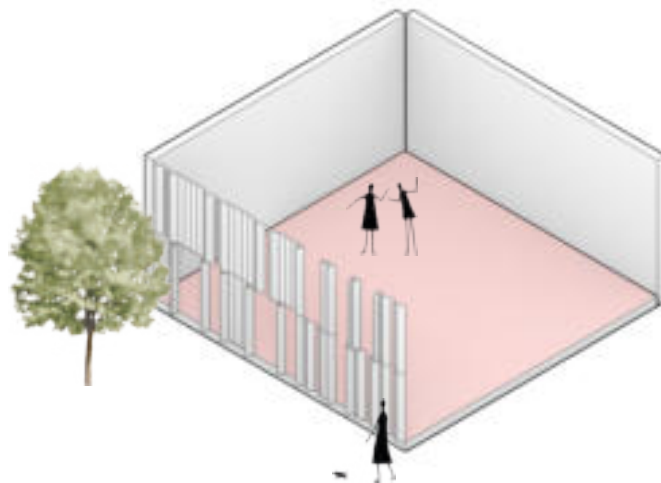
Warehouse

Using partition systems the footprint of the warehouse area expands and contracts based on seasonal demand.



Public space

When the inventory in the warehouse is low, the unused part of the building can be utalized for the public by opening art exhibitions, markets, events, etc



Facade system

The transformative facade allows the structure to change between being closed off and in "warehouse mode" and open up to allow the public to enjoy the building.

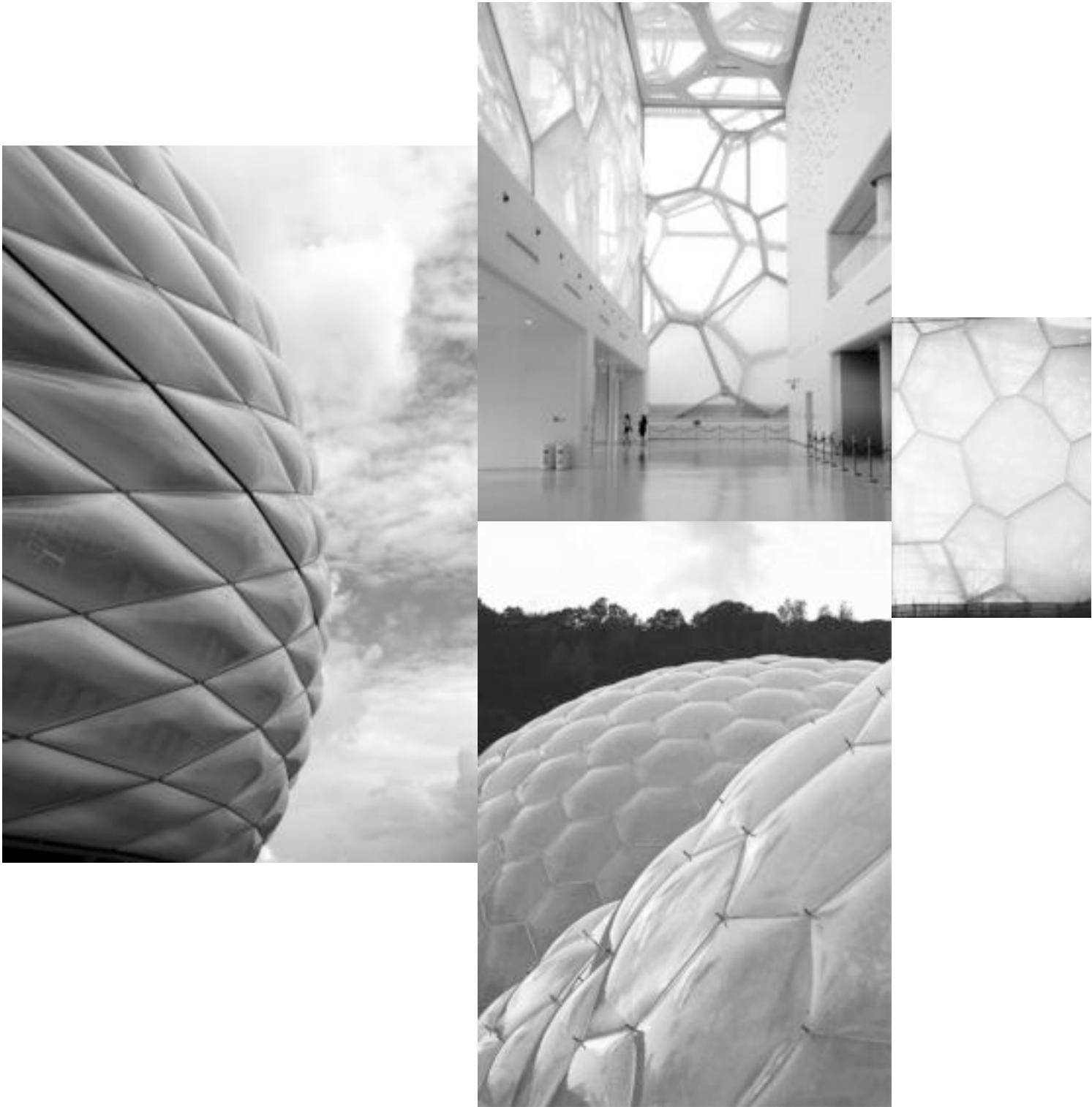
“The warehouse no longer operates as a fixed industrial object, but as a responsive civic infrastructure capable of expanding, contracting, and adapting to changing urban conditions.”

Pneumatic structures

To further reduce the environmental impact of the project, the warehouse envelope transitions from a conventional steel roof system into a lightweight pneumatic ETFE structure. Rather than maintaining a permanently oversized and material-intensive enclosure, the building expands through temporary inflatable volume activated only during periods of increased logistical demand.

ETFE (Ethylene Tetrafluoroethylene) cushions consist of multiple layers of air-filled foil membranes capable of creating highly lightweight yet thermally efficient enclosures. By replacing large portions of the traditional steel envelope with pneumatic systems, the project significantly reduces embodied carbon while minimizing unnecessary permanent building mass.

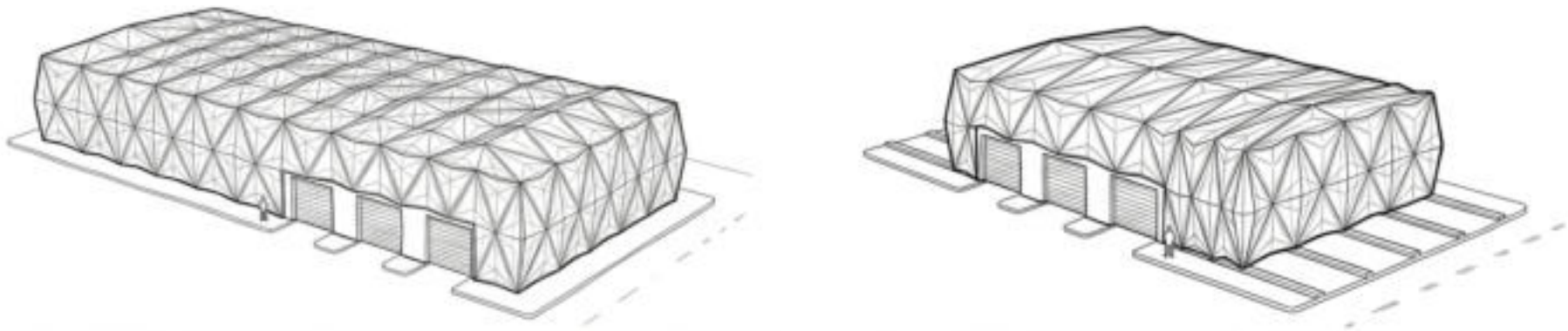
The translucent membrane transforms the warehouse from an opaque industrial container into a naturally daylit interior environment. In Iceland's winter conditions, where daylight is limited for much of the year, access to natural light becomes both an environmental and social sustainability strategy. Increased daylight penetration reduces operational lighting demand while improving circadian health, spatial quality, and employee well-being within the logistics workspace.



Two pneumatic strategies are explored: a fully inflatable envelope and a hybrid system where only the roof structure expands. While the fully pneumatic model offers maximum spatial flexibility, the hybrid approach balances adaptability, structural stability, and environmental performance.

The pneumatic system therefore operates as temporary atmospheric infrastructure; allowing the architecture to expand, contract, and respond dynamically to changing seasonal and operational conditions.





Measured Spatial & Environmental Impact

Key Performance Indicators (KPIs) are simple measurements used to evaluate how well a project performs. In this project, the KPIs compare the proposed design to a traditional warehouse, showing how adaptive systems can improve energy use, flexibility, public space, and environmental performance.

KPI: Adaptive Civic Reallocation

Traditional warehouses often maintain large inactive interior volumes throughout off-peak seasons. By compressing logistical operations into smaller active zones, dormant spaces can temporarily transform into public programs and civic interiors.



KPI: Operational Energy Reduction

Conventional logistics buildings continuously heat, light, and maintain the entire warehouse regardless of occupancy levels. Through adaptive thermal zoning, the project reduces the amount of space requiring active conditioning.



KPI: Visual impact

Shifting from a flat, windowless "grey box" to a more biophilic design to improve neighbor mental well-being.



KPI: Employee working environment

Prioritizing occupant health through Human-Centric Design, the facility incorporates high-performance glazing for daylight autonomy, dedicated restorative resting suites, and direct seamless access to the surrounding green infrastructure.



Green colour indicates the design of the building before

Pink colour indicates the proposed design

All calculations are conceptual estimates used to compare.



Reflection

At the beginning of this semester, logistics was something I had never really thought about. To me, it existed as a hidden and purely functional infrastructure; anonymous warehouses placed at the edges of cities, disconnected from everyday public life. Through this project, however, I began to understand the enormous complexity behind logistics systems and how deeply they influence the way contemporary cities operate.

Researching the topic revealed how many layers of planning, coordination, technology, and human labor are involved in moving goods through society. It also highlighted how many different professions, systems, and networks contribute to keeping logistics functioning continuously in the background of daily life.

What became especially eye-opening was recognizing the environmental impact of these infrastructures. Logistics does not only shape how we consume and receive products, but also affects energy use, urban development, transportation patterns, and environmental performance on a much larger scale than I had previously understood.

Throughout the semester, the project gradually evolved from simply designing a warehouse into questioning how logistics architecture itself could become more adaptive, efficient, and integrated within the city. This process fundamentally changed the way I view industrial infrastructure; not as static background buildings, but as active systems capable of shaping both urban life and environmental futures.

Appendix

This section includes additional research, sketches, and explorations that informed the project but were not included in the final portfolio. These pieces were either personally interesting to me or connected to the research and development behind the work presented throughout the portfolio.

During my case study i also was deeply interested in the problem of urban intergration and how logistic centers can negatively affect neighbourhoods.

During a lecture from Simon from Henning Larsen he said something i found very inspiring; *"will the lady that lives in Lelystad thats walking her dog be happy to walk past the building"*

I spent majority of the semester looking into this topic.

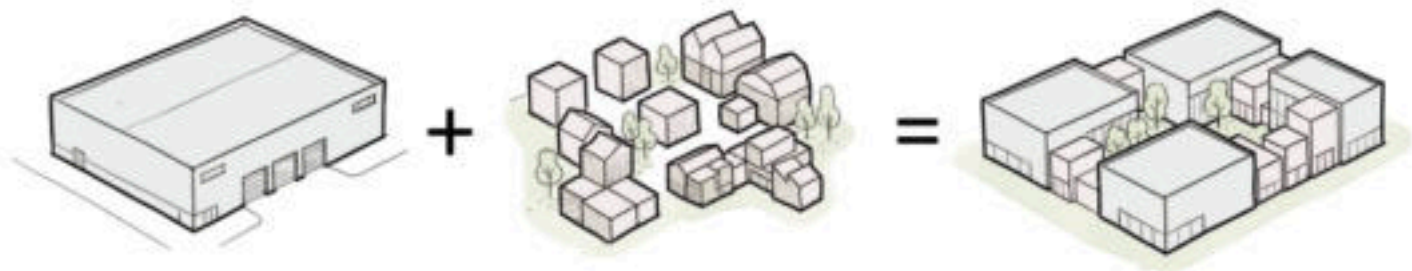
The Challenge

The planning department justified the warehouse as "commercial use," but in reality, it is industrial infrastructure masquerading as urban fabric. This friction, between the massive, blank green walls and the high-density residential properties, creates the primary challenge for this project:

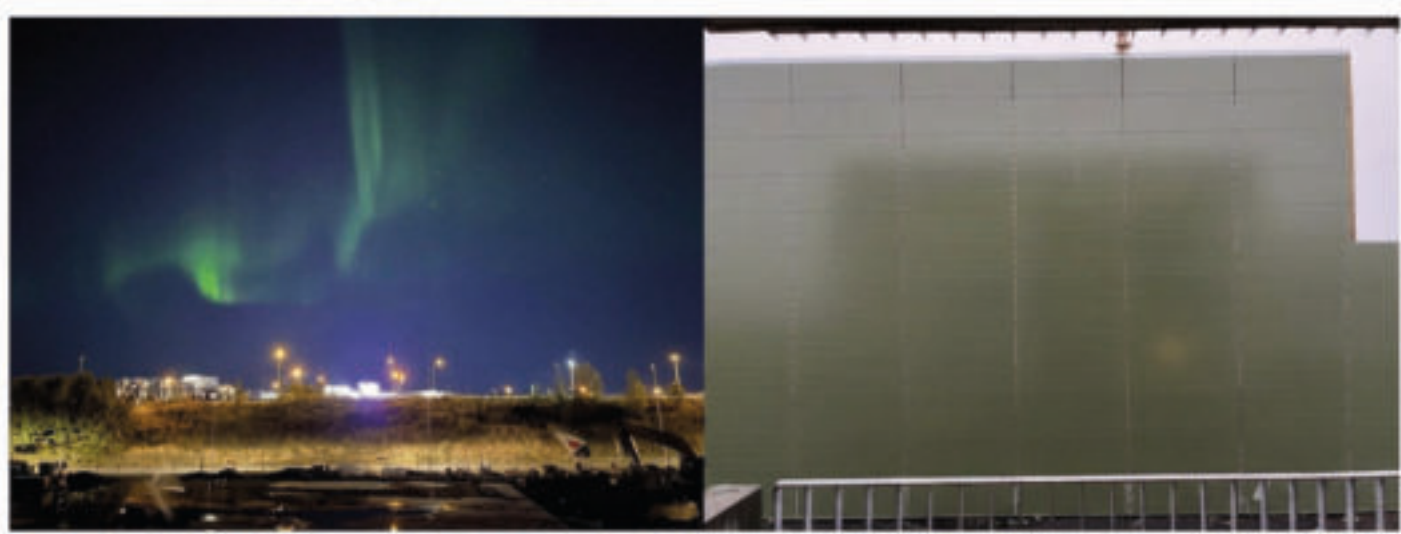
How do we intergrade a logistics buildings into a living neighborhood?

The "Græna gimaldið" stands as a literal wall between the city's ambitions and the community's reality. To move from a "monstrosity" back to a "neighborhood," the intervention must do more than hide the steel, it must dismantle the industrial barrier.

Elastic logistics reimagines the site by breaking apart the monolithic mass into a more open, flexible, and human-scaled environment. Instead of functioning as a single rigid volume, the design dissolves the structure into interconnected elements that create not only movement, but a stronger relationship with the surrounding urban fabric.



Page from previous portfolio work



Photographs from a neighbour living in the nearest residential building from the green monstrosity. Photograpgs show the before and after views from the third floor livingroom window

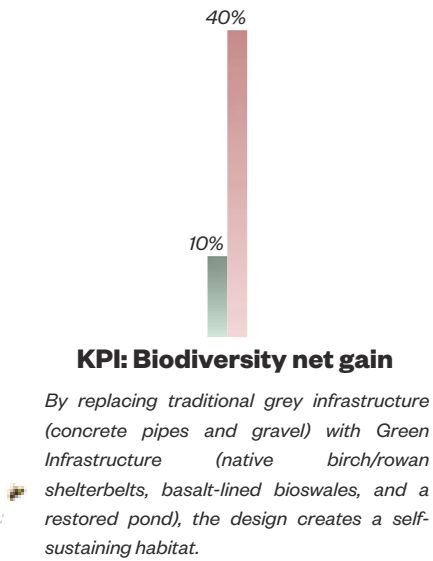


Biodiversity

Stormwater Management: Reykjavík experiences frequent and often intense precipitation. Incorporating sustainable urban drainage systems, such as biodiverse **swales**, helps absorb and naturally manage runoff, reducing pressure on municipal infrastructure while minimizing the risk of site flooding.

Microclimate Regulation: Large warehouse structures can contribute to the formation of urban heat islands. Native vegetation helps regulate surface temperatures, provides natural wind protection, and may contribute to lower heating demands for the facility.

Carbon Sequestration: High-density soil systems combined with moss and native shrub planting significantly improve carbon sequestration, capturing and storing far more carbon than conventional surfaces such as gravel or asphalt while also enhancing the site's ecological value.



A - Betula pubescens



B - Achillea millefolium



C - Salix phylicifolia



D - Sorbus aucuparia



E - Eriophorum vaginatum

Birch (*Betula pubescens*) is Iceland's only native forest-forming tree. It hosts over 100 species of invertebrates, which in turn feed the birds. Provides nesting and food for the Redpoll and the Goldcrest.

Many birds rely on Rowan (*sorbus aucuparia*) for its berries to survive the early winter

A young, healthy Icelandic birch/rowan plantation absorbs approximately **1.5 to 2.0 tonnes of CO2 per hectare per year**.

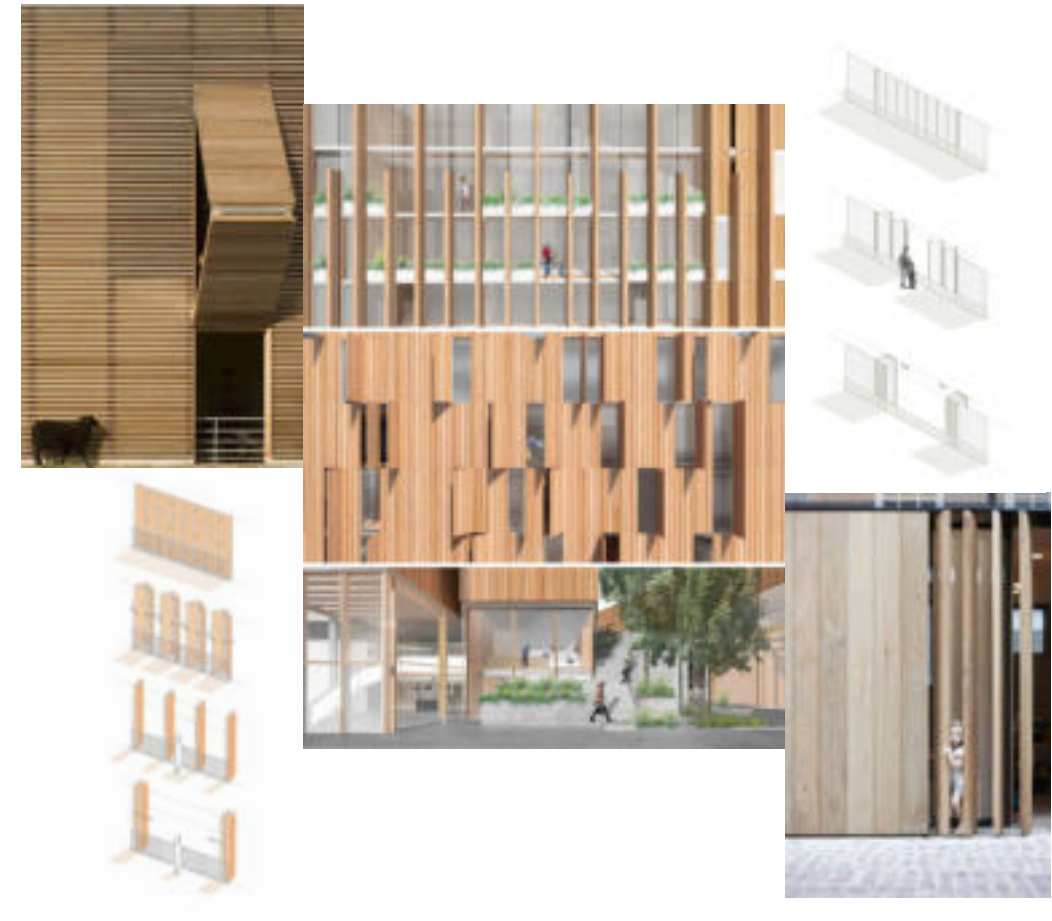
Wetlands containing species like Cottongrass (*Eriophorum vaginatum*) can store significantly more soil carbon than dry land, sometimes exceeding **40% carbon concentration** in the soil compared to just **0.5-2%** in standard industrial soils.

Planting these species of local growing trees and plants an increase from <2 species (baseline) to 8-10 species is achieved, representing a **400%-500% increase** in pollinator variety.





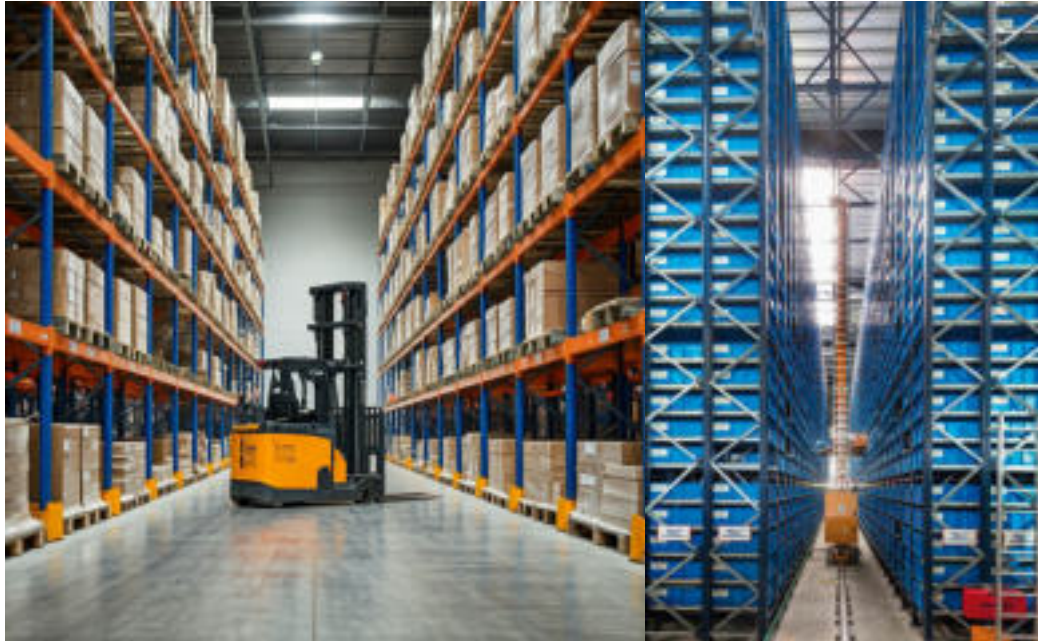
Different **facade strategies** were explored to evaluate varying levels of thermal performance, public permeability, daylight access, and operational flexibility. The selected openable facade system provided the strongest balance between environmental control and civic openness while allowing the building to transition between industrial and public modes.



Different **roof geometries** were evaluated in relation to daylight access, thermal performance, structural efficiency, and environmental adaptability. While flat and sawtooth roof systems provided conventional solutions, the pneumatic roof system offered the highest degree of flexibility, reduced structural weight, and increased daylight diffusion.



Alternative **storage systems (racking systems)** were compared based on space efficiency, operational speed, labor requirements, and adaptability. Robotic storage systems demonstrated higher spatial efficiency and logistical precision, while conventional racking systems remained simpler and more cost-effective.



Both robotic and manually operated **partition systems** were tested as strategies for adaptable thermal zoning and spatial reconfiguration. While robotic offered flexibility and faster reconfiguration, the manual system offered simplicity and lower cost, as well as lower environmental impact.

	System	Reconfiguration Time	Estimated Operational...	Labor Requirement	
1.	Manual Partitions	2-4 hours	Low	High	
2.	Robotic Partitions	5-15 minutes	Medium	Low	
+					

